Astrophysical Variability & Superflares - A Review

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Superflares on Sun-like stars

Schaefer's paper focuses on nine so-called superflares, observed on stars very similar to the Sun.

What superflares are? How defined observationally

Superflares are highly energetic, short-duration flares observed on main-sequence type stars like the Sun. They have energies in the range 10^{33} to 10^{38} ergs, which is orders of magnitudes greater than the energies of ordinary flares (10^{2} - 10^{6} ergs).

A 10^{35} erg (or larger) superflare could severely disrupt life if directed in the direction of Earth at 1 AU. The event would have enough energy to strip away the ozone layer. It would also ionize the atmosphere

The stars where these types of flares have been observed seem very similar to the Sun, spectral-wise.

Proposed explanations

The authors note that known mechanisms that explain ordinary flares are insufficient to explain superflares.

The Sun

The paper gives a rough recurrence rate for a solar-type star on the order of a few decades to centuries, with large uncertainties. As far as our Sun itself is concerned, there are no superflares in the historical record going back to the last 2000 years.

Uncertainty and detection limits

The exact physical mechanism driving such flares on Sun-like stars remains unknown. Most of the evidence comes from long duration photometric surveys, which make it possible to detect rare outbursts like these which would otherwise be missed.

Conclusion

Studying and understanding these specific phenomena is important because

- They are highly energetic whose causes are not well understood
- They seem to be occurring in stars like the Sun, so a threat to life on Earth

In papers like this one, the consensus often leans to one of or a combination of magnetic activity, starspots (sunspots, in the case of the Sun) and rapid rotation for the exact cause, which still eludes scientists. For the Sun in particular, it is statistically likely for a superflare to occur anytime in the near future. The problem is predicting when they will occur, and how often over a specific time period as that would be very relevant to life on Earth. The absence of such an event in the last 2000 years does not rule out anything.

R Coronae borealis stars

R Coronae borealis stars are supergiant stars deficient in Hydrogen and rich in Carbon - high ratios of C-12/C-13. They are very rare in thee galaxy

Observation details

Brightness can drop to ~8 magnitudes in weeks, with status quo being a gradual process taking months or even years. The main cause is because of the formation of carbon dust shells in the star’s outer layers, which obscure the light from the photosphere

Most of these stars pulsate over a period of 40 - 100 days.

Formation theories

There are two leading hypotheses on how these kinds of stars formed:

- Merger of two white dwarfs
- Helium-shell flash in a AGB star

A helium flash is a fusion process where large quantities of helium is converted into carbon in low mass stars like a AGB (asymptotic giant branch) star via the triple-alpha process (see below). In low mass stars, the hydrogen left in the red giant phase is not enough to counteract against the gravitational pressure to do fusion. In some of these stars, if there is sufficient quantities of helium, a runaway fusion process takes place.

The process can generate energy many orders of magnitude greater than the star's previous lifetime output in just a few seconds.

In the merger scenario, we have

- a carbon-oxygen white dwarf
- a helium white dwarf

They are part of a binary system. Once and if their separation gets small enough, the lower mass helium white dwarf is tidally disrupted by its heavier neighbor. The next step, which is the merger itself, is not well grounded in theory, but a few things need to have happened which we can guess. The helium dwarf gets shredded, probably after a gradual process, into a disk enveloping around the carbon-oxygen star. And then, matter begins to accrete into it. The accreted helium is compressed while its temperature and pressure continually gets increased; this triggers the triple-alpha process - burning of helium.

As in helium flash, this same process converts helium into carbon.

Conclusion

Both these hypotheses explain the unusual chemical compositions of these stars, but more evidence is required. But few scientists lean towards the merger model since it

explains the hydrogen deficiency. Even before the merger, white dwarf stars do not have much hydrogen as they lost it earlier in their evolution.

Hydrogen deficiency can be explained away in the helium flash model as well, but the merge model gives a clear reasoning. R coronae borealis stars present a gap in our understanding of stellar evolution. We only have models which do not fully account for their behavior, or how they will evolve in the future.

References

Clayton, Geoffrey C. "The R Coronae borealis stars." Publications of the Astronomical Society of the Pacific 108.721 (1996): 225-225.

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